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BUSINESS INTELLIGENCE

PROJECT PHASE 3

VISUALIZATION & DASHBOARD

Title:

Retail Strategy: The Impact of GDP and Urbanization on European Sales

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1.0 Dashboards

1.1 Dashboard 1 Sales Performance Overview

Dashboard 1 provides a high-level summary of the retail company's overall sales performance across European markets. As shown in Figure 1, the dashboard is designed to allow executives to quickly understand revenue, profitability, and market distribution immediately, while also supporting interactive exploration through three slicers that included Country, Channel Preference, and GDP Tier.

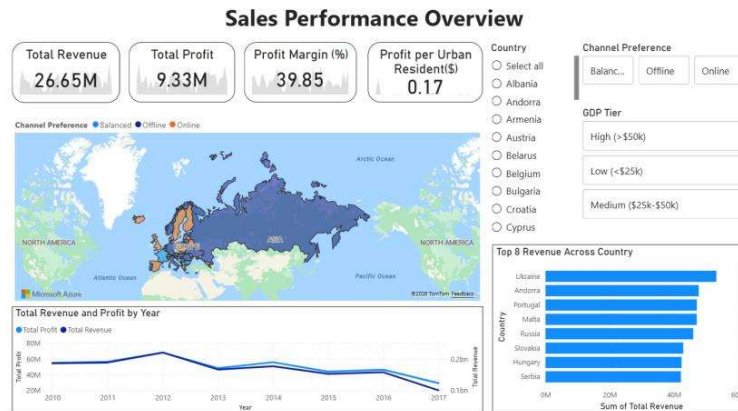


Figure 1 Sales Performance Overview Dashboard

At the top of the dashboard, four KPI cards display the most critical business metrics that includes Total Revenue, Total Profit, Profit Margin, Profit per Urban Resident for all or selected countries. These cards update dynamically based on the slicer selections, allowing users to instantly compare any individual country's performance against the overall baseline.

The filled map in the centre visualizes the geographic distribution of sales across Europe, with each country shaded and color-coded by Channel Preference. The orange representing Online-leaning markets, dark blue for Offline-leaning, and light blue for Balanced markets. This allows the business to immediately identify regional shopping behaviour patterns across the continent.

The line chart at the bottom tracks Total Revenue and Total Profit trends from 2010 to 2017, revealing that sales peaked around 2012 before experiencing a gradual decline toward 2017. This downward trend suggests the company may need to revisit its market strategy in certain regions to reverse the declining performance.

Finally, the Top 8 Revenue Across Country bar chart on the right provides a permanent ranking of the highest contributing markets, with Ukraine, Andorra, and Portugal leading in total revenue. This chart is configured to remain static regardless of slicer selections, serving as a fixed benchmark that users can reference while exploring individual country data through the slicers.

1.2 Dashboard 2 Economic & Urbanization Impact Analysis

As shown in Figure 2, the Economic & Urbanization Impact Analysis dashboard was developed to examine how GDP per capita and urbanization influence retail performance and shopping behaviour across European countries.

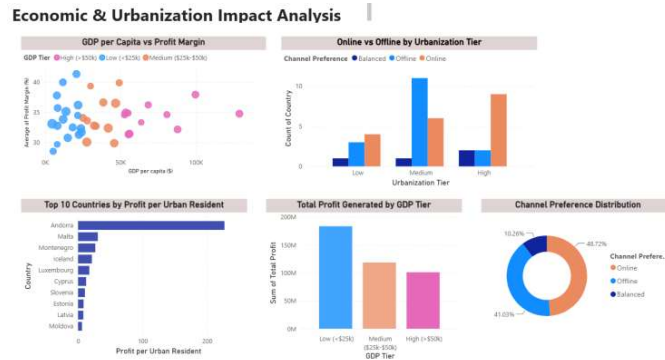


Figure 2 Economic & Urbanization Impact Analysis Dashboard

The **GDP per Capita vs Profit Margin** scatter plot shows the relationship between a country's economic status and profitability. The analysis indicates that countries with higher GDP per capita do not necessarily achieve higher profit margins, suggesting that profitability is influenced by factors beyond economic wealth alone.

The **Online vs Offline by Urbanization Tier** chart compares shopping preferences across different urbanization levels. The results show that highly urbanized countries tend to have a stronger online shopping preference, while medium urbanization countries show a higher offline preference. This suggests that urbanization plays a role in consumer purchasing behaviour.

The **Top 10 Countries by Profit per Urban Resident** chart identifies the most efficient markets based on profit generated relative to urban population size. Countries such as Andorra and Malta demonstrate strong performance and may represent attractive opportunities for future business expansion.

The **Total Profit Generated by GDP Tier** chart compares profitability across GDP groups. The results indicate that countries within the Low GDP Tier contribute the highest total profit, followed by Medium and High GDP Tier countries. Finally, the **Channel Preference Distribution** chart summarizes overall shopping behaviour. Online shopping represents the largest proportion at 48.72%, followed by Offline shopping at 41.03%, while Balanced preferences account for 10.26%.

Overall, the dashboard shows that urbanization has a stronger influence on shopping behaviour than GDP per capita. Businesses should focus on strengthening online retail strategies in highly urbanized markets while also exploring profitable opportunities in lower GDP countries that continue to generate strong profit performance.

2.0 Storyboard

Figure 3 presents the storyboard developed to summarize the key insights obtained from the previous two dashboards. The storyboard highlights the relationship between GDP, urbanization and retail performance across European countries. The first visualization shows that shopping channel preferences vary across Europe, with some countries favouring online retail while others rely more on offline channels. The second visualization demonstrates that higher GDP does not necessarily result in higher profit as countries within the Low GDP tier generate the highest total profit. The third visualization indicates that highly urbanized countries tend to show stronger online shopping preferences compared to less urbanized markets. Finally, the market efficiency analysis identifies countries such as Andorra, Malta and Montenegro as high-performing markets based on profit per urban resident. Overall, the

storyboard provides a clear summary of the factors influencing retail performance and supports data-driven business expansion decisions.

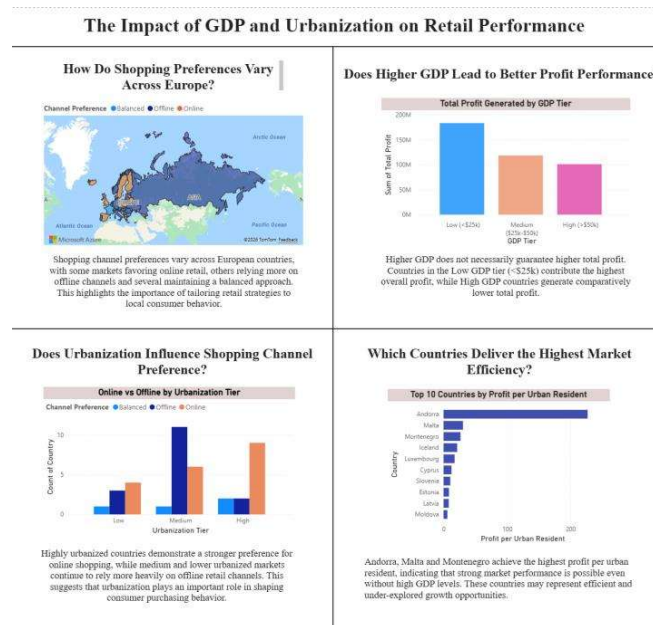


Figure 3 Storyboard showing The Impact of GDP and Urbanization on Retail Performance

3.0 Recommendations

- Expand E-Commerce Investment in Highly Urbanized Markets**
 Countries with high urbanization levels generally demonstrate a stronger preference for online shopping channels. Therefore, the company should continue investing in e-commerce platforms and digital marketing campaigns strategies to maximize market sales performance.
- Maintain Physical Retail Presence in Offline-Oriented Markets**
 Several countries continue to show a stronger preference for offline shopping. Maintaining physical stores and localized retail strategies can help the company strengthen customer relationships and sustain profitability in these markets.
- Prioritize High-Efficiency Emerging Markets**
 Countries such as Andorra, Malta and Montenegro achieve high profit per urban resident despite not belonging to the highest GDP tier. These markets may offer attractive expansion opportunities due to their strong market efficiency and growth potential.

4.0 Key Findings

- Higher GDP does not necessarily result in higher profit performance.
- Highly urbanized markets generally show stronger online shopping preferences.
- Several lower-GDP countries like Andorra, Malta and Montenegro demonstrate strong market efficiency despite not belonging to the highest GDP tier.
- Profit per urban resident is a useful indicator for identifying attractive expansion opportunities.
- Expansion decisions should consider both economic and consumer behaviour factors rather than GDP alone.